


Figure 1: Interaction of a user with a DBMS (Image source: *googleimages.com*)

distributed database.

5. **End user databases:** There is a variety of data available at the workstations of different end users of an organisation. Each workstation acts like a small database in itself, and has data in the form of spreadsheets, presentations, downloaded files, Word files and in Notepad format. All such small databases together form an end user database.

Now let's have a look at some of the popular open source databases available in the market.

## MySQL

MySQL has been in use since 1995 and is currently owned by Oracle. In addition to its open source version, there are a couple of paid editions available as well, which offer some additional features, such as cluster geo-replication and automatic scaling. We all know that MySQL has become an industry standard now, as it is compatible with almost every operating system and has been written in both C and C++. This database option is great for different international users as well, since the server can provide various error messages to the clients in a number of languages.

***Pros:***

- Offers a flexible privilege and password system
- Can be used even if there is no available network
- Uses host-based verification
- It consists of different libraries that can be embedded into various standalone applications
- It has security encryption for all the password traffic
- Supports servers as a separate program for the client server networked environment

***Cons:***

- Users feel that MySQL no longer falls under the free OS category
- It is no longer community driven, so bug fixes and patches do not happen in time
- Falls behind other similar available options due to its slow updates

## PostgreSQL

PostgreSQL, also called Postgres, is basically an object-relational DBMS with much emphasis on extensibility and standards compliance. As a database server, its important functions are to securely store data and return the data or data sets in response to different requests made from other software applications. PostgreSQL can handle different workloads, ranging from that of small single-machine applications to even large Internet-facing applications with many concurrent users. It has been developed by PostgreSQL Global Development Group, and is available as free and open source software.

***Pros:***

- It is transactional and ACID (atomicity, consistency, isolation, durability)-compliant
- Supports updatable views and materialised views
- Functions, stored procedures, triggers, etc, can be very well used in it
- Supports concurrency with the help of a system known as MVCC (Multi Version Concurrency Control).
- Provides three levels of transaction isolation, which are: Read Committed, Repeatable Read and Serialisable
- Supports serialisability using the serialisable snapshot isolation technique
- Supports a large variety of data types like Boolean, binary, etc

***Cons:***

- It does not have any bug tracker (whereas it supports a bug-submission form, which feeds into the pgsql-bugs mailing list), making it a bit difficult to know the status of bugs.

## MariaDB

MariaDB is a database that is widely used by tech giants like Wikipedia, Facebook and even Google. It has been developed by various developers who worked as part of MySQL. It is basically a database server that provides drop-in replacement functionality for MySQL. Data security is one of the most significant concerns and priorities for developers of MariaDB, and almost in each of its solution releases the developers merge in all of MySQL's security patches and also enhance the same, if required.

***Pros:***

- It provides real-time access to the data sets
- It supports the maximum number of core functionalities of MySQL (it is an alternative to MySQL)
- It supports high scalability with easier integration
- It provides a couple of alternate patches, storage engines and server optimisations